

CS & IT ENGINEERING



Computer Network

Introduction

Lecture No. - 06

By - Abhishek Sir





Recap of Previous Lecture



Topic

Data Link Layer

Topic

Physical Layer

Topic

Network Topology



Topics to be Covered



Topic

Network Topology

Topic

Switching

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : Network Topology

→ Arrangement of hosts inside a network.

→ Different types of topology are :

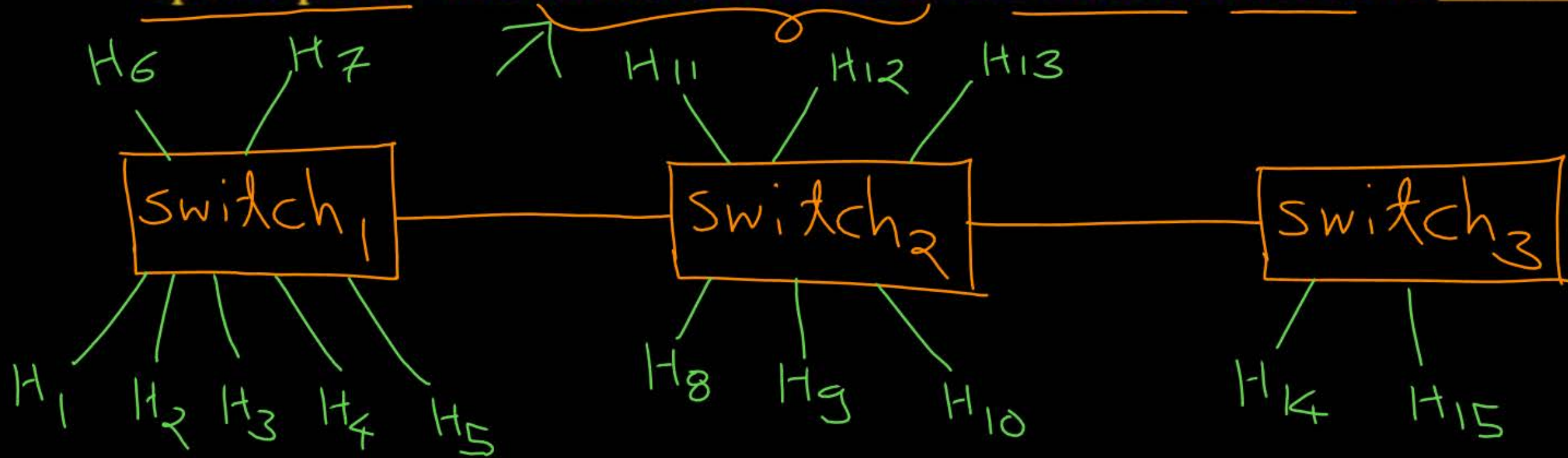
1. Mesh
2. Star
3. Bus
4. Ring

[NAT]

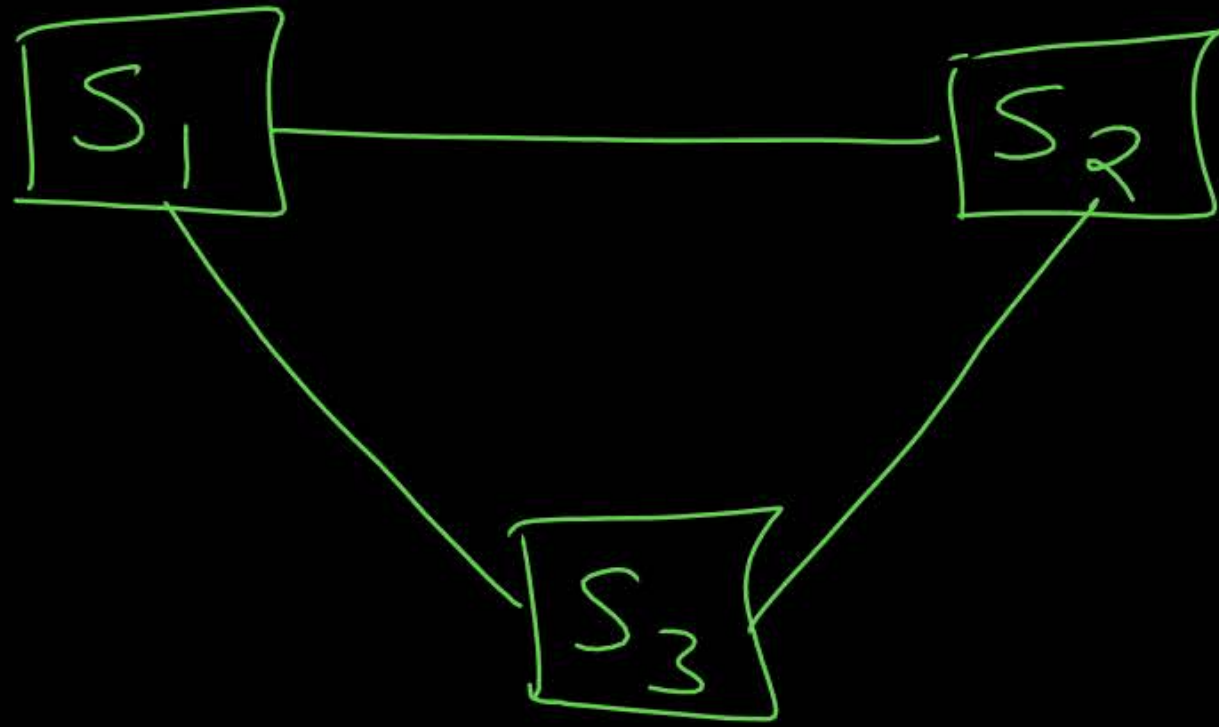
[GATE-2019][2 Mark]



#Q. Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____.



Ans = 3





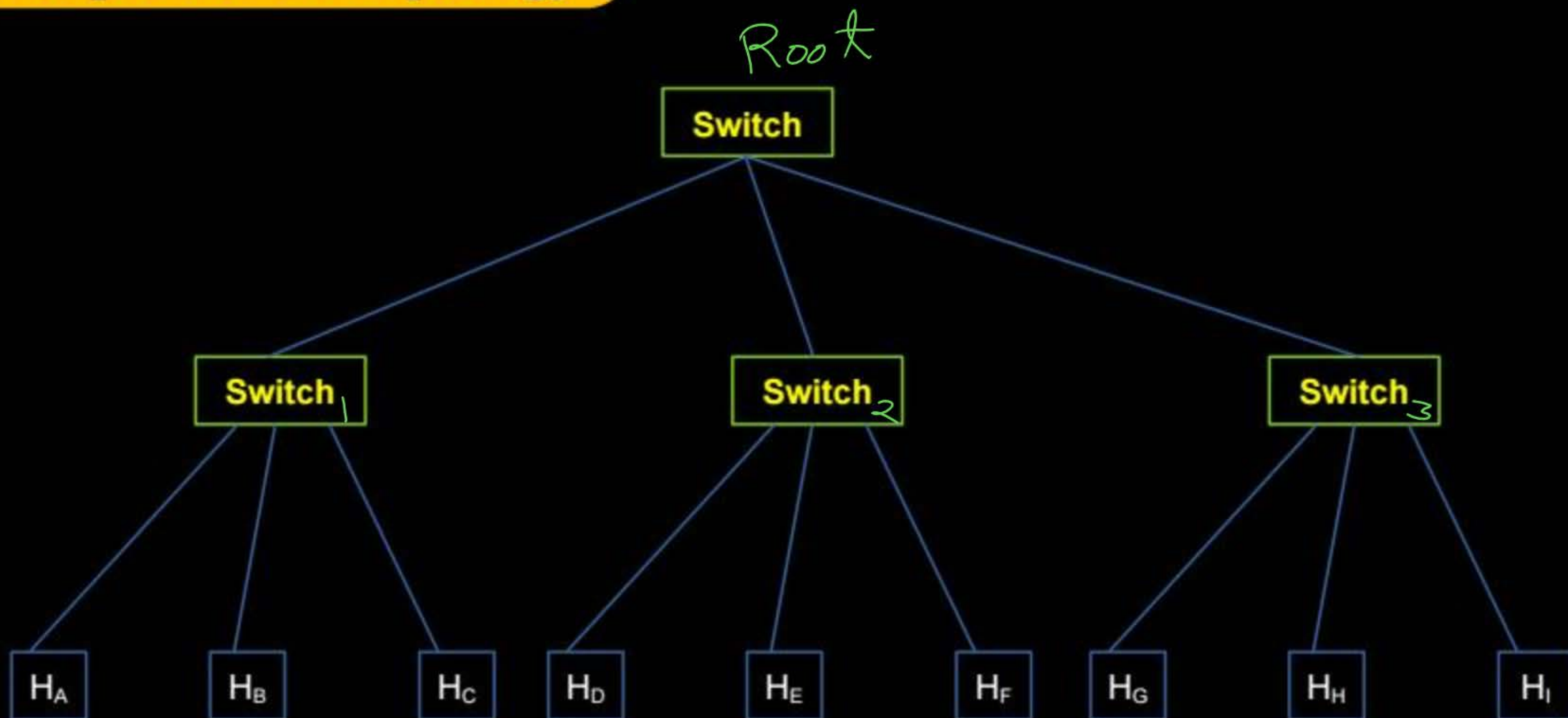
Topic : Tree Topology



- hierarchical structure
[with a central root node]
- Extension of the star topology ✓
- Networking devices (like switches) often have "uplink" ports



Topic : Tree Topology





Topic : Uplink Port



- Special type of Ethernet port
[To connect network devices like switches or routers]
- Often designed with higher bandwidth than regular ports
[To handle larger amounts of data traffic]



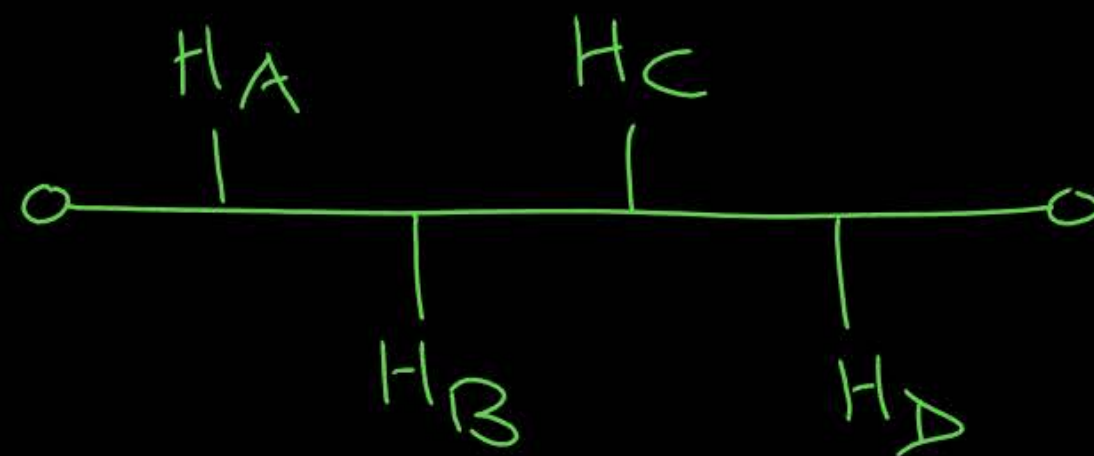
Topic : Bus Topology



→ Every host connected to centralized backbone media (coaxial cable).

Total number of nodes = n

Total number of links = 1 ✓



→ Multipoint [Multidrop]

→ Access Control Method

[Channel Access Protocols] → MAC Sublayer

[ALOHA, CSMA] → CSMA/CD



Topic : Bus Topology



Advantages :-

- Installation cost is very low
[Preferable for long area network]

Disadvantages :-

- If backbone media fails
then all communication stop



Topic : Ring Topology



→ Every host connected to two adjacent hosts using **point-to-point link** in cyclic fashion.

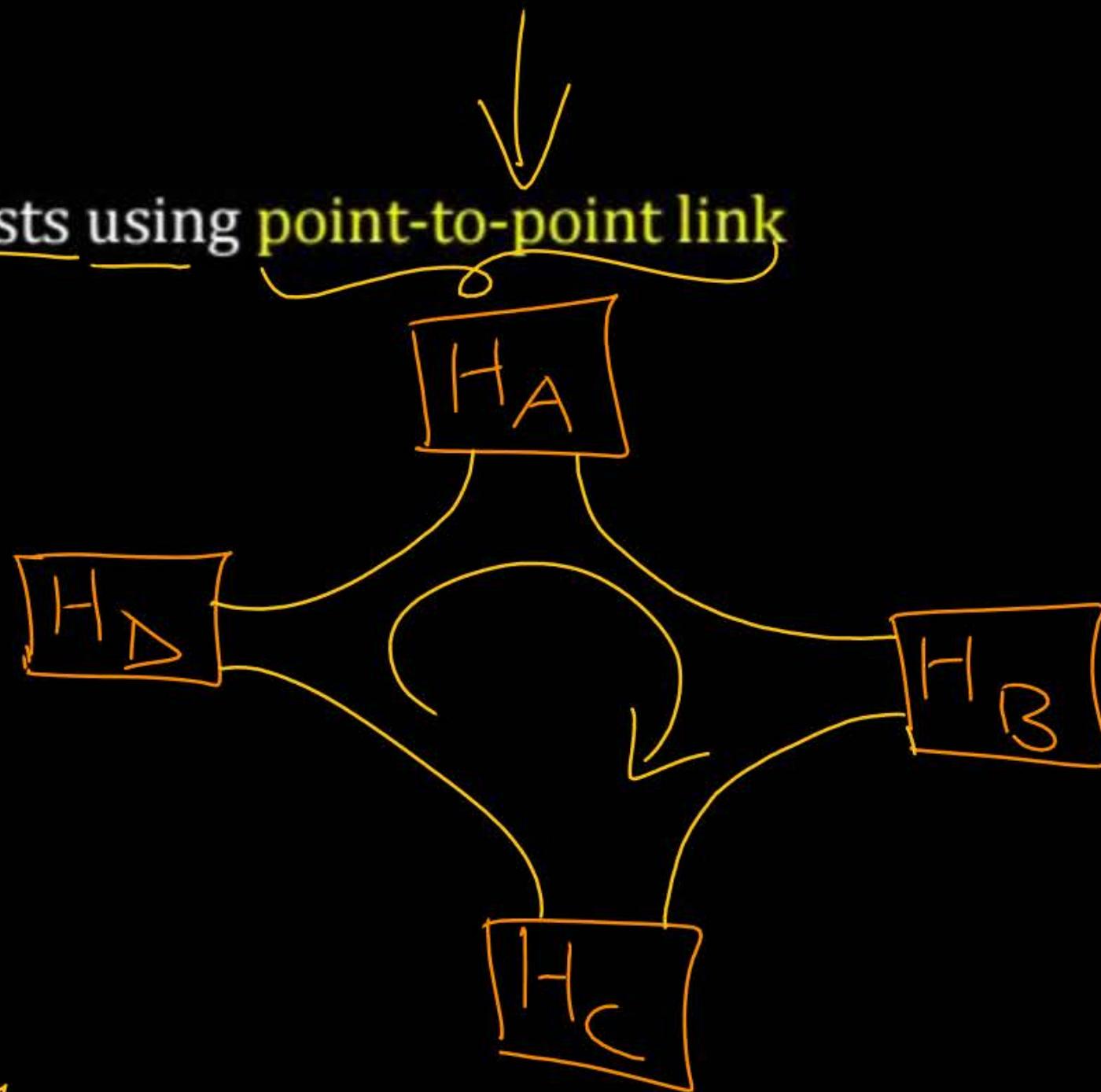
Total number of nodes = n

Total number of links = n

→ **Simplex Communication**

→ Access Control Method

↓
Token ⇒ Token Ring





Topic : Syllabus



Concept of layering : OSI and TCP/IP Protocol Stacks;

(Introduction)

Basics of packet, circuit and virtual circuit switching;

↓
↓
(Imp.)



Topic : Switching



→ Process to move data (or packets) towards destination over the network

→ Types of switching techniques:

1. Circuit Switching

2. Packet Switching

- 1) Datagram Network (Packet Switching)
- 2) Virtual circuit switching



Topic : Circuit Switching



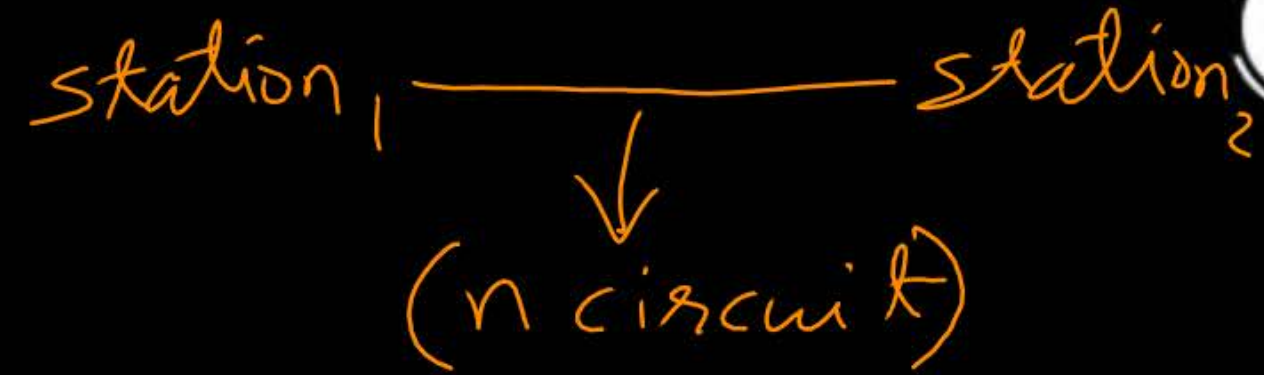
→ Establishes dedicated circuit between sender and receiver, before transmission
[Over the links of the network]

→ Phases of Circuit Switching :

1. Circuit establishment ✓
2. Data transfer
3. Circuit disconnect

→ Example : Telephone Networks

[PSTN : Public Switched Telephone Network]



=> Circuit in a link implemented with

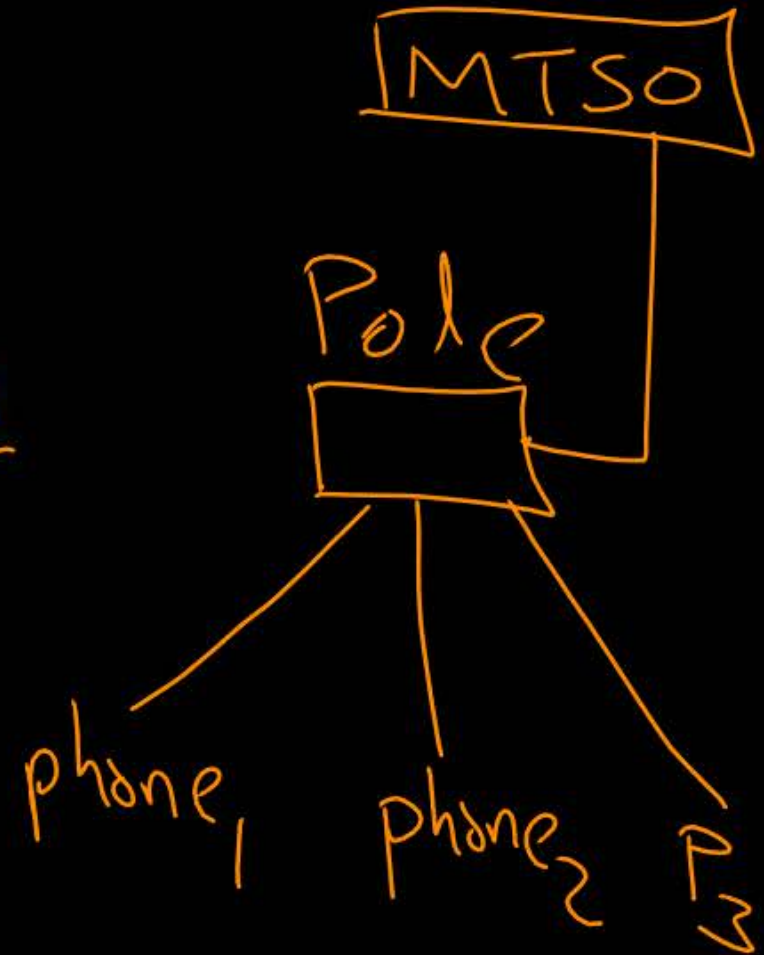
1. Frequency Division Multiplexing [FDM]

- Frequency spectrum of a link
- Analog circuit switching

2. Time Division Multiplexing [TDM]

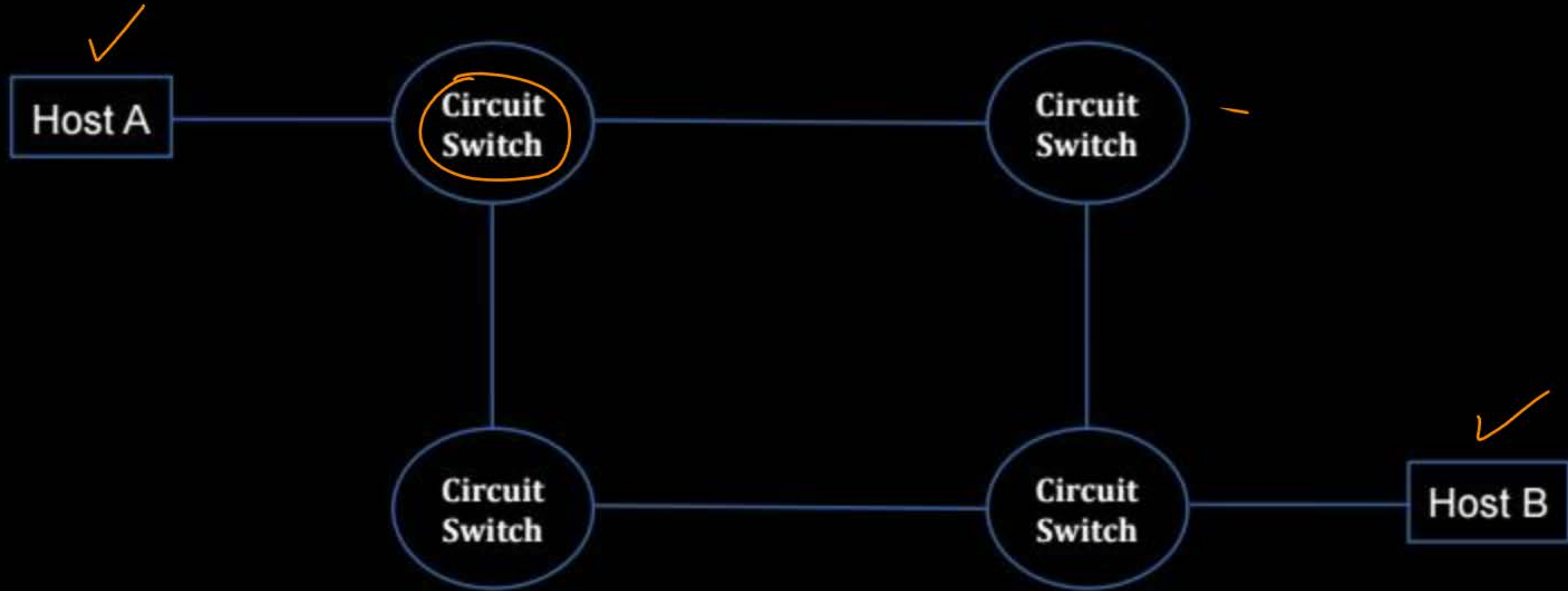
- Time is divided into frames of fixed duration
- Each frame is divided into fixed number of time slots
- Digital circuit switching

=> Each link contains n "circuits" [TDM or FDM]





Topic : Circuit Switching





Topic : Circuit Switching



Advantage :-

→ Congestion may occur during circuit establishment

[No any congestion occur, during data transfer]

→ All data (or packets) follow each other on reserved path

[Data (or packets) having same end-to-end delay]

A

B



Topic : Circuit Switching

(V. Imp.)



Disadvantage :-

- Inefficient utilization of network resources (in compare to packet switching)
- Expensive



Topic : Message Switching

→ Application processes doing communication by exchanging "messages"

Advantage :-

- No any dedicated path required between sender and receiver [unlike circuit switching]
- Store and Forward →

Disadvantage :-

- Entire message is transmitted as single unit



Topic : Packet Switching ✓



→ Message is divided into smaller size packets
[Packets may be same or different size]

→ Example : Internet is a packet-switched network

Optimum Packet Size → Flow Control



2 mins Summary



Topic

Network Topology ✓

Topic

Switching

- 1) Basics of Circuit switching ✓
- 2) Packet Switching

[Extra class: 10 AM]



THANK - YOU

